Jet-Controlled Tokenomics for a Solana Land NFT Economy: A Preliminary Dynamic and Stochastic Model

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June 11, 2026

Abstract

This paper proposes a preliminary mathematical model for a blockchain-based land economy implemented on Solana. The proposed system, called *Solana Land NFT*, combines land ownership, resource production, marketplace exchange, staking, renting, guild competition, and adaptive token issuance. The central object is a game economy whose state is described by a vector of endogenous variables including active players, land activity, resources, token supply, token price, staking, and utility. We describe the game mechanics, token flows, nonlinear dynamic equations, stochastic components, and a jet-based control layer for adaptive tokenomics. The main idea is that the economy should not react only to current values such as price or circulating supply, but also to higher-order local trends: growth, acceleration, deceleration, and instability signals. This leads to a control mechanism based on finite jets of economic observables.

1 Introduction

Blockchain games often suffer from unstable token economies. A typical failure mode occurs when rewards are emitted faster than in-game utility and token sinks can absorb them. This produces inflation, speculative extraction, declining prices, and eventually player churn. The purpose of this article is to outline a mathematical framework for a land-based NFT economy in which token issuance, burning, rewards, and marketplace incentives are governed by dynamic and stochastic feedback mechanisms.

The proposed game, called *Solana Land NFT*, is a land-based Web3 strategy economy. Players own or rent land parcels represented by NFTs, extract resources, construct buildings, upgrade land, trade assets, participate in guilds, and use the native token as the fuel of the economy.

The core mathematical object is the state vector

$$Y(t) = \begin{pmatrix} P(t) \\ L(t) \\ R(t) \\ S(t) \\ T(t) \\ X(t) \\ U(t) \end{pmatrix},$$

where

$$P(t) = \text{active players},$$

$$L(t) = \text{active land parcels},$$
$$R(t) = \text{aggregate resources},$$
$$S(t) = \text{circulating token supply},$$
$$T(t) = \text{staked token amount},$$
$$X(t) = \text{token price},$$
$$U(t) = \text{economic utility of the game}.$$

The central hypothesis is that a sustainable token economy should be controlled not only by

$$Y(t),$$

but by its finite jet

$$J^k Y(t) = \big(Y(t), Y'(t), Y''(t), \ldots, Y^{(k)}(t)\big).$$

2 The Game Economy

2.1 Land NFTs

The game world contains a finite number of land parcels:

$$N_{\text{land}} = 100\,000.$$

Each land parcel is represented by an NFT

$$\text{Land}_i.$$

A land parcel is described by a vector

$$\text{Land}_i = (z_i, r_i, q_i, b_i, \ell_i, a_i),$$

where

$$z_i = \text{zone}, \qquad r_i = \text{rarity}, \qquad q_i = \text{productivity},$$
$$b_i = \text{buildings}, \qquad \ell_i = \text{upgrade level}, \qquad a_i = \text{activity score}.$$

The rarity classes are

$$\mathcal{R} = \{\text{Common}, \text{Rare}, \text{Epic}, \text{Legendary}\}.$$

A possible distribution is

Rarity	Number of parcels	Productivity multiplier
Common	60 000	1
Rare	25 000	1.5
Epic	10 000	2.5
Legendary	5 000	5

The land productivity is modeled as

$$q_i = q_0 m_r(i) m_z(i) m_\ell(i),$$

where

$$m_r(i) = \text{rarity multiplier},$$
$$m_z(i) = \text{zone multiplier},$$
$$m_\ell(i) = 1 + \rho\sqrt{\ell_i}.$$

2.2 Map as a Graph

The world map is modeled as a finite graph

$$G = (V, E),$$

where

$$V = \{\mathrm{Land}_1, \ldots, \mathrm{Land}_N\}$$

is the set of land parcels and

$$E \subseteq V \times V$$

is the set of adjacency relations.

If

$$(\mathrm{Land}_i, \mathrm{Land}_j) \in E,$$

then land parcels i and j are neighbors. The graph distance is denoted by

$$d_G(i, j).$$

The transportation cost between two land parcels is

$$C_{\mathrm{transport}}(i, j) = c_0 + c_1 d_G(i, j).$$

Let A be the adjacency matrix:

$$A_{ij} = \begin{cases} 1, & i \sim j, \\ 0, & i \not\sim j. \end{cases}$$

The graph Laplacian is

$$\mathcal{L} = D - A.$$

Resource diffusion across the map may be modeled by

$$\frac{d\mathbf{R}}{dt} = -\mathcal{L}\mathbf{R} + \mathbf{Q} - \mathbf{C},$$

where

$$\mathbf{R} = \text{resource vector}, \qquad \mathbf{Q} = \text{production vector}, \qquad \mathbf{C} = \text{consumption vector}.$$

3 Resources and Production

Let

$$\mathcal{Z} = \{\mathrm{Forest}, \mathrm{Mine}, \mathrm{Desert}, \mathrm{Water}, \mathrm{Crystal}, \mathrm{City}\}$$

be the set of land zones.

Each zone produces different resources. For example:

$$\mathcal{R}_{\mathrm{res}} = \{W, S, O, C, M, G\},$$

where

$$W = \text{wood}, \quad S = \text{stone}, \quad O = \text{ore}, \quad C = \text{crystal}, \quad M = \text{mana}, \quad G = \text{gold}.$$

For land parcel i, let

$$\mathbf{r}_i(t) = (r_{i,1}(t), \ldots, r_{i,m}(t))$$

be its resource vector.

The deterministic production of land i is

$$Q_i(t) = q_i a_i(t),$$

where

$$a_i(t) \in [0,1]$$

is the activity score.

A stochastic production model is

$$Q_i(t) = q_i a_i(t) + \sigma_i \xi_i(t),$$

where

$$\xi_i(t) \sim N(0,1).$$

Rare resource drops are modeled by Bernoulli random variables:

$$Y_i(t) \sim \text{Bernoulli}(p_i).$$

If

$$Y_i(t) = 1,$$

then a rare resource is generated.

4 Buildings and Upgrades

Players may construct buildings on land parcels:

$$\mathcal{B} = \{\text{Farm}, \text{Mine}, \text{Workshop}, \text{Market}, \text{Temple}, \text{Defense}\}.$$

A building acts as a transformation of the land parameter vector:

$$T_B : \Theta \to \Theta.$$

If

$$\theta_i = (q_i, p_i, f_i, m_i),$$

then

$$\theta_i \mapsto T_B(\theta_i).$$

Examples:

$$q_i \mapsto q_i(1 + \alpha_F)$$

for a farm,

$$p_i \mapsto p_i + \Delta p$$

for a mine,

$$f_i \mapsto f_i(1 - \rho)$$

for a market.

The upgrade level is

$$\ell_i \in \{0, 1, \ldots, 10\}.$$

The upgrade cost is nonlinear:

$$C_\ell = C_0 a^\ell, \qquad a > 1.$$

The productivity increase is sublinear:

$$q_i(\ell) = q_i(0)(1 + \rho\sqrt{\ell}).$$

This asymmetry,

$$C_\ell \sim a^\ell, \qquad q_i(\ell) \sim \sqrt{\ell},$$

prevents explosive productivity growth.

5 Native Token and Token Allocation

Let the native utility token be denoted by

$$\text{LAND}.$$

The maximum supply is

$$S_{\max} = 1\,000\,000\,000.$$

A preliminary token allocation is:

Category	Share	Tokens
Play-to-earn rewards	30%	300 000 000
Staking rewards	20%	200 000 000
Team	15%	150 000 000
Treasury/DAO	15%	150 000 000
Liquidity	10%	100 000 000
Investors/partners	7%	70 000 000
Airdrop/community	3%	30 000 000

The token is used for

$$\text{upgrade}, \quad \text{crafting}, \quad \text{renting}, \quad \text{marketplace}, \quad \text{staking}, \quad \text{governance}.$$

6 Emission and Burning

The basic emission schedule is exponentially decreasing:

$$E(t) = E_0 e^{-\alpha t}, \qquad E_0 > 0, \qquad \alpha > 0.$$

The total burn is

$$B(t) = B_{\text{upgrade}}(t) + B_{\text{craft}}(t) + B_{\text{market}}(t) + B_{\text{rent}}(t) + B_{\text{travel}}(t).$$

The circulating supply satisfies

$$S(t+1) = S(t) + E(t) - B(t).$$

In continuous time:

$$\frac{dS}{dt} = E(t) - B(t).$$

A simple activity-driven burn model is

$$B(t) = \beta c P(t),$$

where

$$\beta = \text{average burn rate}, \qquad c = \text{average daily fee per player}.$$

Thus

$$\frac{dS}{dt} = E_0 e^{-\alpha t} - \beta c P(t).$$

A sustainable long-run economy should satisfy

$$\limsup_{t\to\infty} \big(E(t) - B(t)\big) \le 0.$$

7 Reward Allocation

Let

$$E_{\text{land}}(t) = \gamma_L E(t)$$

be the land activity reward pool, with

$$0 < \gamma_L < 1.$$

The reward of land parcel i is

$$\text{Reward}_i(t) = E_{\text{land}}(t) \frac{q_i a_i(t)}{\sum_{j=1}^{N} q_j a_j(t)}.$$

Hence

$$\sum_{i=1}^{N} \text{Reward}_i(t) = E_{\text{land}}(t).$$

This mechanism prevents passive NFT ownership from extracting the reward pool without contribution.

8 Rental System

A rental contract is represented by

$$\text{Rent}_{i,u} = (\text{Land}_i, u, p, \tau),$$

where

$$u = \text{renter}, \qquad p = \text{rental price}, \qquad \tau = \text{rental duration}.$$

The produced yield is divided as

$$Y_i = Y_i^{\text{renter}} + Y_i^{\text{owner}} + Y_i^{\text{burn}} + Y_i^{\text{treasury}}.$$

A possible split is

$$\begin{aligned} Y_i^{\text{renter}} &= 0.70 Y_i, \\ Y_i^{\text{owner}} &= 0.20 Y_i, \\ Y_i^{\text{burn}} &= 0.05 Y_i, \\ Y_i^{\text{treasury}} &= 0.05 Y_i. \end{aligned}$$

9 Marketplace

If an asset is sold for value

$$V,$$

the marketplace fee is

$$F = fV.$$

For example,

$$f = 0.025.$$

The fee split is

$$\begin{aligned} F_{\text{burn}} &= 0.4F, \\ F_{\text{treasury}} &= 0.4F, \\ F_{\text{creator}} &= 0.2F. \end{aligned}$$

The marketplace therefore functions as both a liquidity mechanism and a token sink.

10 Guilds and Regional Competition

Let

$$\mathcal{G} = \{G_1, \ldots, G_m\}$$

be the set of guilds. A guild G_k controls a set of land parcels

$$\mathcal{L}_k \subseteq V.$$

The power of guild G_k is

$$\Pi_k(t) = \sum_{i \in \mathcal{L}_k} q_i(1 + \ell_i)a_i(t).$$

If guilds compete for a regional reward $R_z(t)$, then the reward of guild G_k is

$$\text{Reward}_{G_k}(t) = R_z(t)\frac{\Pi_k(t)}{\sum_{j=1}^{m} \Pi_j(t)}.$$

A probabilistic competition model is

$$\mathbb{P}(G_k \text{ wins at time } t) = \frac{\Pi_k(t)}{\sum_{j=1}^{m} \Pi_j(t)}.$$

11 Deterministic Dynamic System

The aggregate economy may be described by the discrete-time system

$$\begin{cases} P(t+1) = P(t) + A(t) - C(t), \\ L(t+1) = L(t) + L_{\text{on}}(t) - L_{\text{off}}(t), \\ R(t+1) = R(t) + Q(t) - D_R(t), \\ S(t+1) = S(t) + E(t) - B(t), \\ T(t+1) = T(t) + T_{\text{in}}(t) - T_{\text{out}}(t), \\ X(t) = k\dfrac{D_{\text{LAND}}(t)}{S(t) - T(t) + \varepsilon}. \end{cases}$$

Here

$$P(t) = \text{active players},$$

$$L(t) = \text{active land parcels},$$

$$R(t) = \text{resources},$$

$$S(t) = \text{circulating supply},$$

$$T(t) = \text{staked supply},$$

$$X(t) = \text{token price}.$$

The effective circulating supply is

$$S_{\text{eff}}(t) = S(t) - T(t).$$

The demand for the token is modeled as

$$D_{\text{LAND}}(t) = c_1P(t) + c_2U(t) + c_3M(t),$$

where

$$M(t) = \text{marketplace volume}.$$

Thus

$$X(t) = k\frac{c_1P(t) + c_2U(t) + c_3M(t)}{S(t) - T(t) + \varepsilon}.$$

12 Nonlinear System

A more explicit nonlinear model is

$$P(t+1) = P(t) + \lambda_0 + \lambda_1 X(t) + \lambda_2 U(t) - p_c P(t).$$

Utility is defined by

$$U(t) = w_1 M(t) + w_2 Q(t) + w_3 R_{\text{rent}}(t) + w_4 C_{\text{craft}}(t) + w_5 T(t),$$

with

$$w_1 + \cdots + w_5 = 1.$$

Staking evolves according to

$$T(t+1) = T(t) + \eta X(t)(S(t) - T(t)) - \theta T(t).$$

Burn is modeled by

$$B(t) = \beta_1 P(t) + \beta_2 M(t) + \beta_3 U(t).$$

The supply equation becomes

$$S(t+1) = S(t) + E_0 e^{-\alpha t} - \beta_1 P(t) - \beta_2 M(t) - \beta_3 U(t).$$

Definition 1. *The economy is called* long-run anti-inflationary *if*

$$\limsup_{t\to\infty} \frac{S(t+1) - S(t)}{S(t)} \le 0.$$

Definition 2. *The economy is called* utility-supported *if token demand satisfies*

$$D_{\text{LAND}}(t) = D_{\text{utility}}(t) + D_{\text{speculation}}(t)$$

and

$$\liminf_{t\to\infty} \frac{D_{\text{utility}}(t)}{D_{\text{LAND}}(t)} > 0.$$

13 Stochastic Components

Player arrivals are modeled by a Poisson process:

$$A(t) \sim \text{Poisson}(\lambda(t)).$$

The intensity may depend on utility and price:

$$\lambda(t) = \lambda_0 + \lambda_1 U(t) + \lambda_2 X(t).$$

Player churn is modeled by a binomial distribution:

$$C(t) \sim \text{Binomial}(P(t), p_c(t)).$$

The churn probability may be decreasing in utility:

$$p_c(t) = p_0 e^{-\gamma U(t)}.$$

Rare item drops are modeled as

$$Y_i(t) \sim \text{Bernoulli}(p_i).$$

Random production shocks are modeled as

$$Q_i(t) = q_i a_i(t) + \sigma_i \xi_i(t), \qquad \xi_i(t) \sim N(0,1).$$

The token price may be modeled by a stochastic differential equation:

$$dX_t = \mu(P_t, R_t, S_t, T_t, U_t) X_t \, dt + \sigma_X X_t \, dW_t.$$

A utility-sensitive drift is

$$\mu(P, R, S, T, U) = a_X U + b_X P - c_X S + d_X T.$$

Thus

$$dX_t = (a_X U_t + b_X P_t - c_X S_t + d_X T_t) X_t \, dt + \sigma_X X_t \, dW_t.$$

14 Continuous-Time Stochastic Model

The full stochastic economy may be approximated by

$$\begin{cases} dP_t = \left(aP_t \left(1 - \dfrac{P_t}{K} \right) + bU_t - cP_t \right) dt + \sigma_P dW_t^P, \\ dR_t = (\alpha_R L_t - \beta_R P_t R_t - \delta_R R_t) dt + \sigma_R dW_t^R, \\ dS_t = (E_0 e^{-\alpha t} - B_t) dt, \\ dT_t = (\eta X_t (S_t - T_t) - \theta T_t) dt + \sigma_T dW_t^T, \\ dX_t = (a_X U_t + b_X P_t - c_X S_t + d_X T_t) X_t dt + \sigma_X X_t dW_t^X. \end{cases}$$

This is a coupled nonlinear stochastic system. It contains:

logistic growth,

resource depletion,

token emission and burn,

staking feedback,

market price diffusion.

15 Jet-Controlled Tokenomics

The main proposal of this article is to introduce a jet-based control layer.

Definition 3. *Let*

$$Y(t)$$

be the state vector of the economy. The finite jet of order k is

$$J^k Y(t) = \left(Y(t), Y'(t), Y''(t), \ldots, Y^{(k)}(t) \right).$$

If the deterministic part of the economy is

$$Y' = F(Y),$$

then the total derivative along the vector field F is

$$\delta_F = \sum_i F_i(Y) \frac{\partial}{\partial Y_i}.$$

The higher-order derivatives are

$$Y^{(n+1)} = \delta_F^n(F)(Y).$$

This gives an algebraic mechanism for computing local economic trends.

16 Jet of Token Supply

Let

$$S' = E - B,$$

where

$$E(t) = E_0 e^{-\alpha t}, \qquad B(t) = \beta P(t).$$

Then

$$S' = E_0 e^{-\alpha t} - \beta P.$$

The second derivative is

$$S'' = -\alpha E_0 e^{-\alpha t} - \beta P'.$$

If

$$P' = aP\left(1 - \frac{P}{K}\right) - cP,$$

then

$$S'' = -\alpha E_0 e^{-\alpha t} - \beta \left[aP\left(1 - \frac{P}{K}\right) - cP\right].$$

Let

$$F(P) = (a - c)P - \frac{a}{K}P^2.$$

Then

$$P' = F(P),$$
$$P'' = F'(P)F(P).$$

Therefore

$$P'' = \left(a - c - \frac{2a}{K}P\right)\left((a - c)P - \frac{a}{K}P^2\right),$$

and

$$S''' = \alpha^2 E_0 e^{-\alpha t} - \beta \left(a - c - \frac{2a}{K}P\right)\left((a - c)P - \frac{a}{K}P^2\right).$$

17 Jet of Token Price

Let

$$X = k\frac{D}{S - T + \varepsilon}.$$

Assume

$$D = c_1 P + c_2 U.$$

Put

$$N = c_1 P + c_2 U, \qquad M = S - T + \varepsilon.$$

Then

$$X = k\frac{N}{M}.$$

The first derivative is

$$X' = k\frac{N'M - NM'}{M^2}.$$

The second derivative is

$$X'' = k\left(\frac{N''M - NM''}{M^2} - 2\frac{(N'M - NM')M'}{M^3}\right).$$

The signs of

$$X', \qquad X'', \qquad X'''$$

indicate local growth, acceleration, and loss of momentum.

18 Jet-Based Stability Indicators

A normalized jet stability index may be defined as

$$\mathcal{J}(t) = a_1\frac{X'}{X} + a_2\frac{X''}{X} + a_3\frac{U'}{U} - a_4\frac{S'}{S} - a_5\left|\frac{S''}{S}\right|.$$

The economy is locally stable if

$$\mathcal{J}(t) > 0.$$

It is locally unstable if

$$\mathcal{J}(t) < 0.$$

A bubble signal occurs when

$$X' > 0, \qquad X'' > 0, \qquad \frac{U'}{U} \approx 0.$$

A death-spiral signal occurs when

$$X' < 0, \qquad P' < 0, \qquad U' < 0, \qquad S' > 0.$$

19 Adaptive Emission

Instead of a purely exogenous emission schedule

$$E(t) = E_0 e^{-\alpha t},$$

we define a jet-controlled emission function:

$$E(t) = E_0 e^{-\alpha t} C(J^2 Y(t)).$$

A possible correction factor is

$$C(J^2 Y) = \exp\left(-\lambda_1 \frac{S'}{S} - \lambda_2 \frac{X''}{X} + \lambda_3 \frac{U'}{U}\right).$$

If token supply grows too quickly, then

$$\frac{S'}{S} > 0$$

reduces emission. If utility grows, then

$$\frac{U'}{U} > 0$$

allows greater reward distribution.

20 Adaptive Burn

Let

$$B(t) = \beta(t) F(t),$$

where

$$F(t) = \text{total in-game fees.}$$

The adaptive burn rate is

$$\beta(t) = \text{clip}\left(\beta_0 + b_1 \frac{S'}{S} - b_2 \frac{U'}{U} - b_3 \frac{X'}{X}, \beta_{\min}, \beta_{\max}\right).$$

Thus burn increases when supply growth is excessive or price dynamics deteriorate.

21 Adaptive Rewards

The total reward pool is

$$\text{RewardPool}(t) = E_0 e^{-\alpha t} \exp\left(-\lambda \frac{S'}{S} + \mu \frac{U'}{U}\right).$$

The reward of land i is

$$\text{Reward}_i(t) = \text{RewardPool}(t) \frac{q_i a_i(t)}{\sum_j q_j a_j(t)}.$$

This creates a feedback rule:

$$J^k Y(t) \mapsto (E(t), B(t), \text{Reward}(t)).$$

22 Discrete Jet Features for Machine Learning

In practice, economic data is observed in discrete time. The first finite difference is

$$\Delta Y_t = Y_t - Y_{t-1}.$$

The second finite difference is

$$\Delta^2 Y_t = Y_t - 2Y_{t-1} + Y_{t-2}.$$

The third finite difference is

$$\Delta^3 Y_t = Y_t - 3Y_{t-1} + 3Y_{t-2} - Y_{t-3}.$$

The discrete jet feature vector is

$$\Phi_k(t) = \left[Y_t, \Delta Y_t, \Delta^2 Y_t, \ldots, \Delta^k Y_t\right].$$

A predictive model may estimate

$$\mathbb{P}(\text{crash in } h \text{ days} \mid \Phi_k(t))$$

or

$$\mathbb{E}[X(t+h) \mid \Phi_k(t)].$$

23 Main Formal Statements

Theorem 1 (Reward Conservation)**.** *If rewards are distributed according to*

$$\text{Reward}_i(t) = E_{\text{land}}(t)\frac{q_i a_i(t)}{\sum_j q_j a_j(t)},$$

then

$$\sum_i \text{Reward}_i(t) = E_{\text{land}}(t).$$

Theorem 2 (Anti-Inflationary Supply Condition)**.** *If*

$$B(t) \geq E(t)$$

for all sufficiently large t, then the circulating supply is eventually non-increasing:

$$S(t+1) \leq S(t).$$

Theorem 3 (Utility-Supported Price Stability)**.** *Assume*

$$X(t) = k\frac{D_{\text{LAND}}(t)}{S(t) - T(t) + \varepsilon}.$$

If

$$D_{\text{LAND}}(t)$$

is bounded below by utility demand and

$$S(t) - T(t)$$

is bounded above, then $X(t)$ admits a positive lower bound.

Theorem 4 (Jet-Controlled Emission Response). *Let*

$$E(t) = E_0 e^{-\alpha t} \exp\left(-\lambda_1 \frac{S'}{S} - \lambda_2 \frac{X''}{X} + \lambda_3 \frac{U'}{U}\right).$$

Then emission decreases with normalized supply growth and speculative acceleration, and increases with normalized utility growth.

Theorem 5 (Bubble Signal). *If*

$$X' > 0, \qquad X'' > 0, \qquad U' \approx 0,$$

then price acceleration is not supported by utility growth.

Theorem 6 (Death-Spiral Signal). *If*

$$X' < 0, \qquad P' < 0, \qquad U' < 0, \qquad S' > 0,$$

then the system exhibits simultaneous price decline, user decline, utility decline, and inflationary supply pressure.

24 Conclusion

This article proposed a preliminary mathematical framework for a Solana-based land NFT economy. The game combines land ownership, resource production, buildings, upgrades, renting, marketplace exchange, guild competition, and token-governed incentives. The economy is modeled as a nonlinear dynamic system with stochastic player arrivals, churn, resource drops, production shocks, and token price diffusion.

The main contribution is the introduction of jet-controlled tokenomics. Instead of reacting only to current observables such as price, supply, player count, or marketplace volume, the system observes local trends through finite jets:

$$J^k Y(t) = (Y(t), Y'(t), \ldots, Y^{(k)}(t)).$$

This makes it possible to detect bubbles, death spirals, excessive inflation, declining utility, and unstable reward dynamics earlier than with static indicators.

The proposed framework suggests a new class of Web3 economic mechanisms:

$$\text{economic state} \to \text{finite jet} \to \text{adaptive emission, burn, and reward control.}$$

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